

Annie Prasanna Manoharan

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EDUCATION

University of Arkansas

Master of Science in Computer Engineering

Aug 2024 – May 2026

Fayetteville, AR

Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Privacy Enhancing Techniques

Sri Ramakrishna Engineering College

Bachelor of Engineering in Electronics and Communication

Aug 2019 – May 2023

India

TECHNICAL SKILLS

Cloud and Tools: Model monitoring and drift detection, Model deployment and serving, Experiment tracking and versioning, Git, ML pipeline automation, Data curation and annotation workflows, Docker, Linux, Grafana, CI/CD, Jupyter, Data validation and quality

Machine Learning and AI: PyTorch, Computer vision, Scikit-learn, Flask, Pandas, NumPy

Languages: Python, SQL

Data Engineering: ETL/ELT

PROJECTS

MiGa - Multi-Chicken Gait Assessment (AICV Lab)

YOLO11n-Pose, OC-SORT, LSTM, PyTorch

- Integrated YOLO11n-Pose for keypoint estimation, applied OC-SORT for multi-object tracking, and constructed a many-to-one LSTM sequence network for classification
- Built a data engineering framework to preprocess and segment raw frames into a detection split of 4,569 frames with 11,422 bounding boxes and a regression split of 55 expert-annotated sequences
- Implemented spatial normalization into local coordinate spaces to eliminate translational variance, achieving keypoint mAP 91.83% at 1,250 FPS, 0.93 MOTA, and 0.91 IDF1

Real-Time Medical IoT Analytics Platform *Apache Kafka, Apache Flink, Flink SQL, Python, ClickHouse, Grafana*

- Developed an end-to-end, sub-second latency streaming platform using Apache Flink SQL and Apache Kafka to ingest, process, and score high-frequency medical IoT device data
- Implemented temporal Flink OVER windows partitioned by device ID to compute rolling Z-scores over 5-second sliding intervals, applying custom watermark delay strategies to handle out-of-order streams
- Designed ClickHouse MergeTree schemas sorted by composite keys to align disk layout with clinical query patterns, with an optimized Python multi-topic batch consumer isolating ingestion

PROFESSIONAL EXPERIENCE

Data Engineering Intern

InsOps Inc.

Jul 2024 – Present

Chicago, Illinois

- Built data workflows integrating SQL Server, Azure SQL, AWS S3, and cloud databases, preparing structured datasets for downstream data science and analytics
- Developed data pipelines in AWS using AWS Glue, Lambda, EMR, SQL, and Apache Airflow for orchestration, with S3 cloud storage, for scalable data ingestion, transformation, validation, and analytics

Teaching Assistant

University of Arkansas, Dept. of Computer Science and Engineering

Sep 2024 – May 2026

Fayetteville, AR

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals

Research Assistant

AICV Lab, University of Arkansas

Sep 2024 – Jan 2026

Fayetteville, AR

- Built reusable preprocessing workflows to segment, organize, and transform raw image/video data into structured inputs for downstream processing and evaluation
- Evaluated results using mAP, MOTA, IDF1, PSNR, SSIM, and FID

Project Engineer

Soliton Technologies

Jun 2023 – Jun 2024

India

- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation
- Developed customized software solutions for Dräger as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements